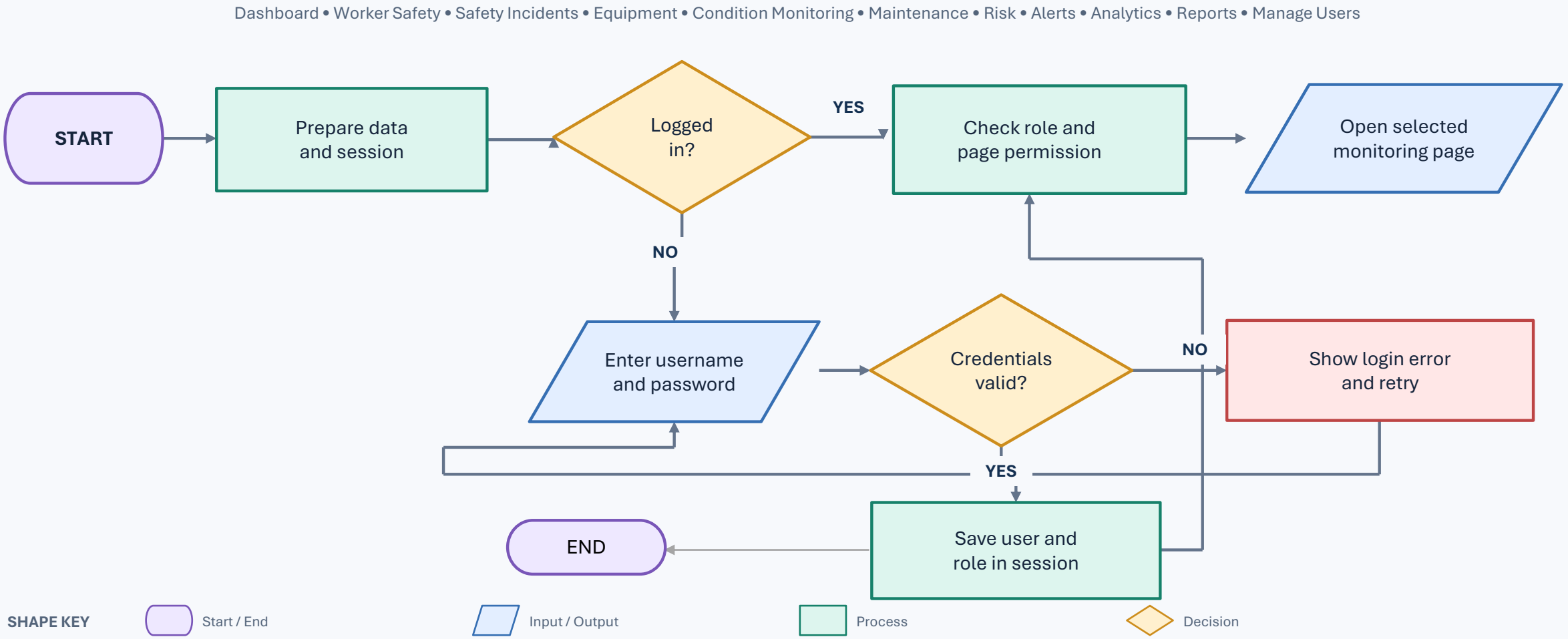
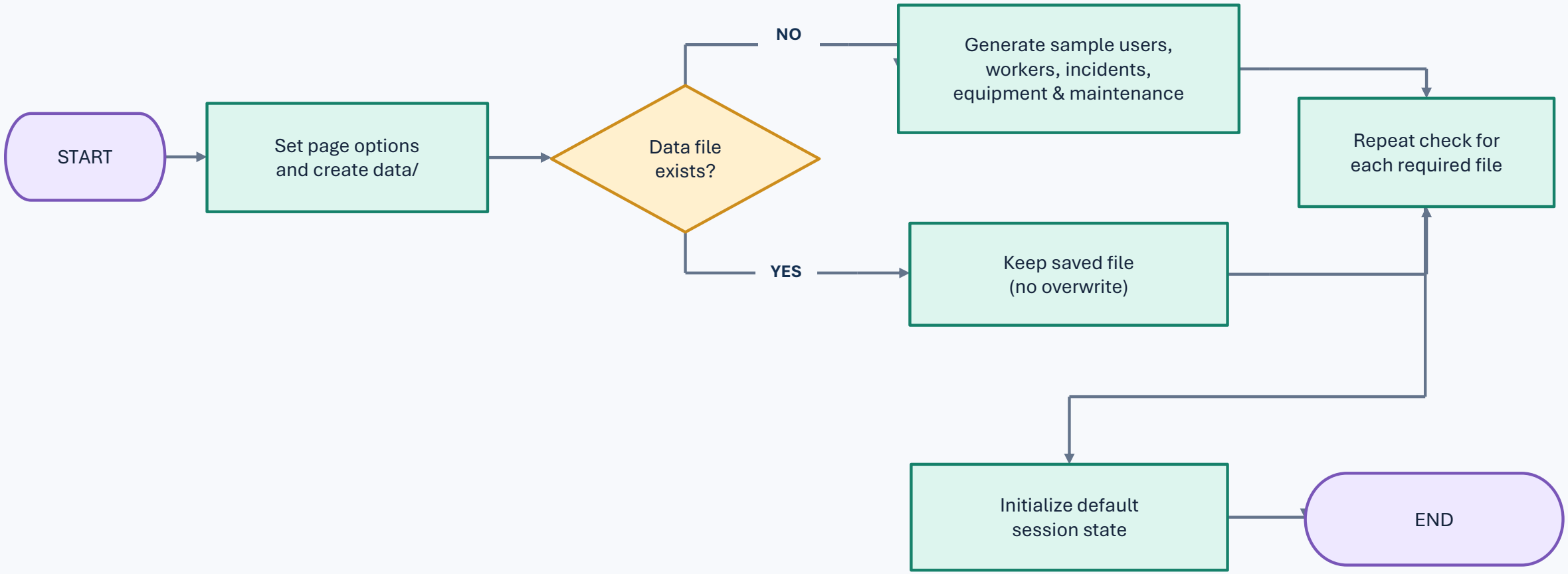
Khwezi Mining Monitor | System overview

How the application starts, signs users in, and opens role-appropriate monitoring pages. Logout resets the session.



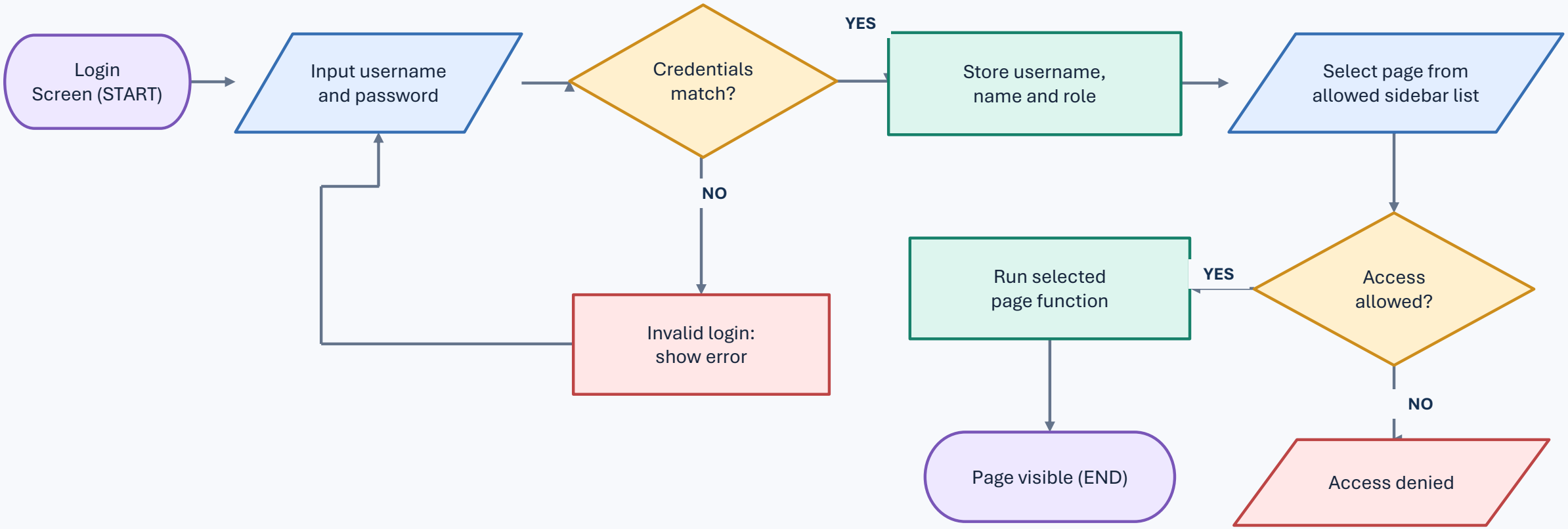
Startup | Prepare application data

On first launch, sample files are generated. On later launches, existing files are reused.



Authentication | Permissions and navigation

The program validates passwords, checks role permissions, and selects a page function.

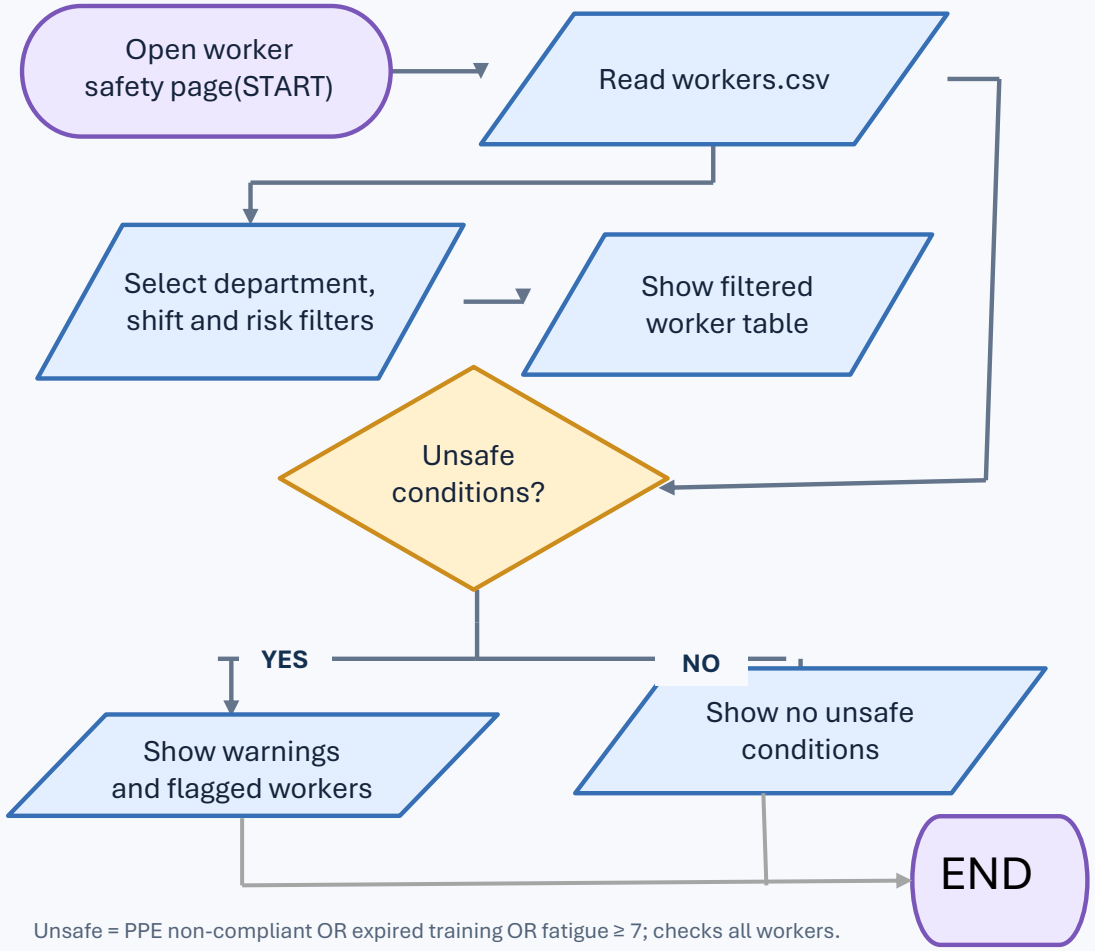


Administrator: all pages • Other roles: only their listed pages • Logout: clear stored session keys, rerun login.

Worker safety | Incident management

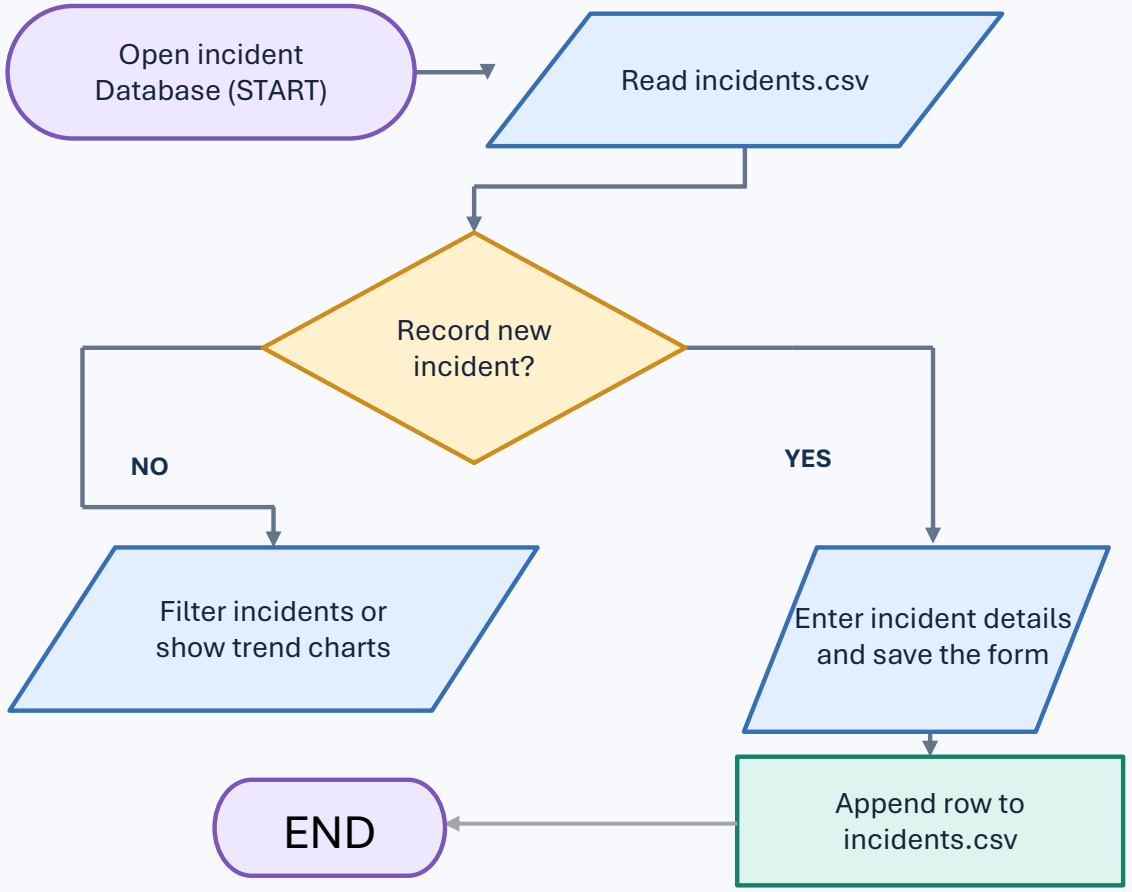
Two connected safety features: review worker exposure and maintain incident records.

A • WORKER SAFETY



Unsafe = PPE non-compliant OR expired training OR fatigue ≥ 7; checks all workers.

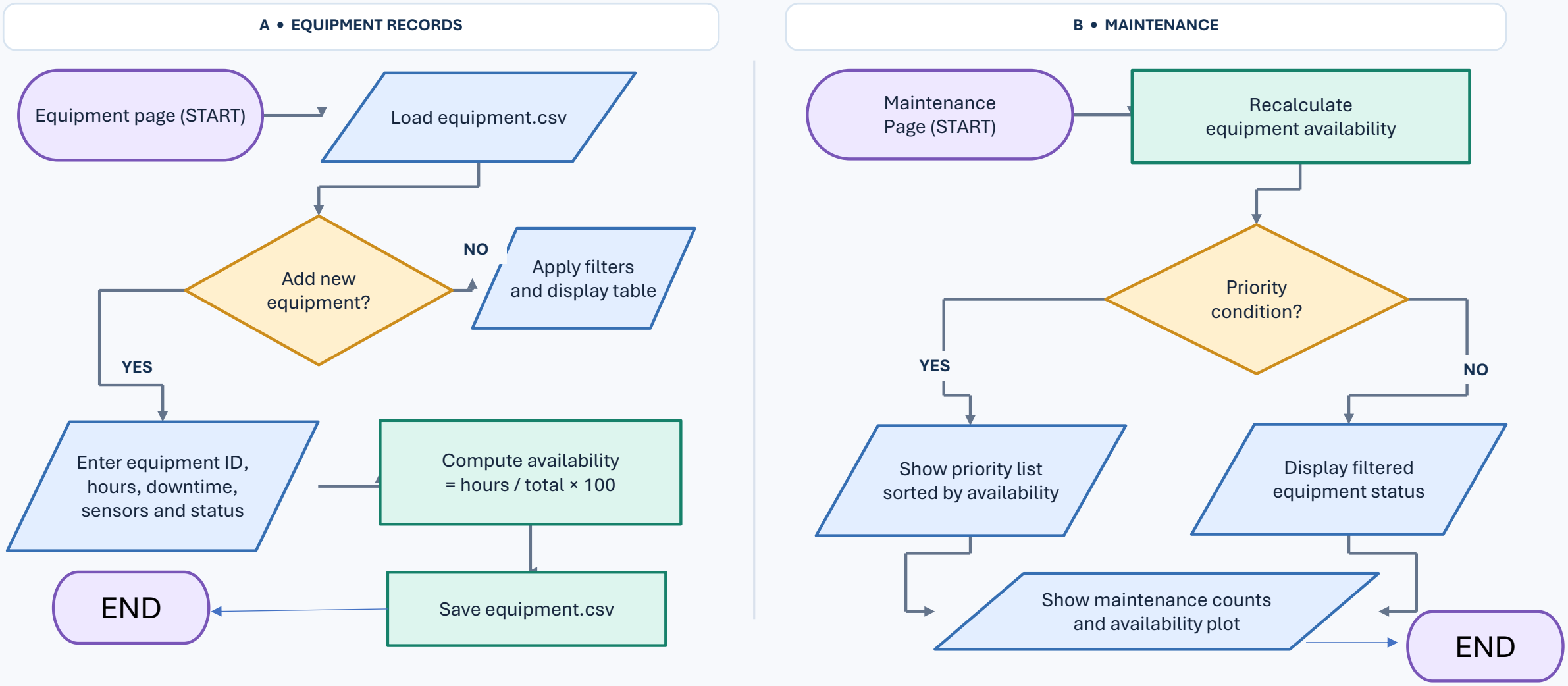
B • INCIDENTS



Incident form saves to CSV; chart tab groups records by incident type, month, shift and cause.

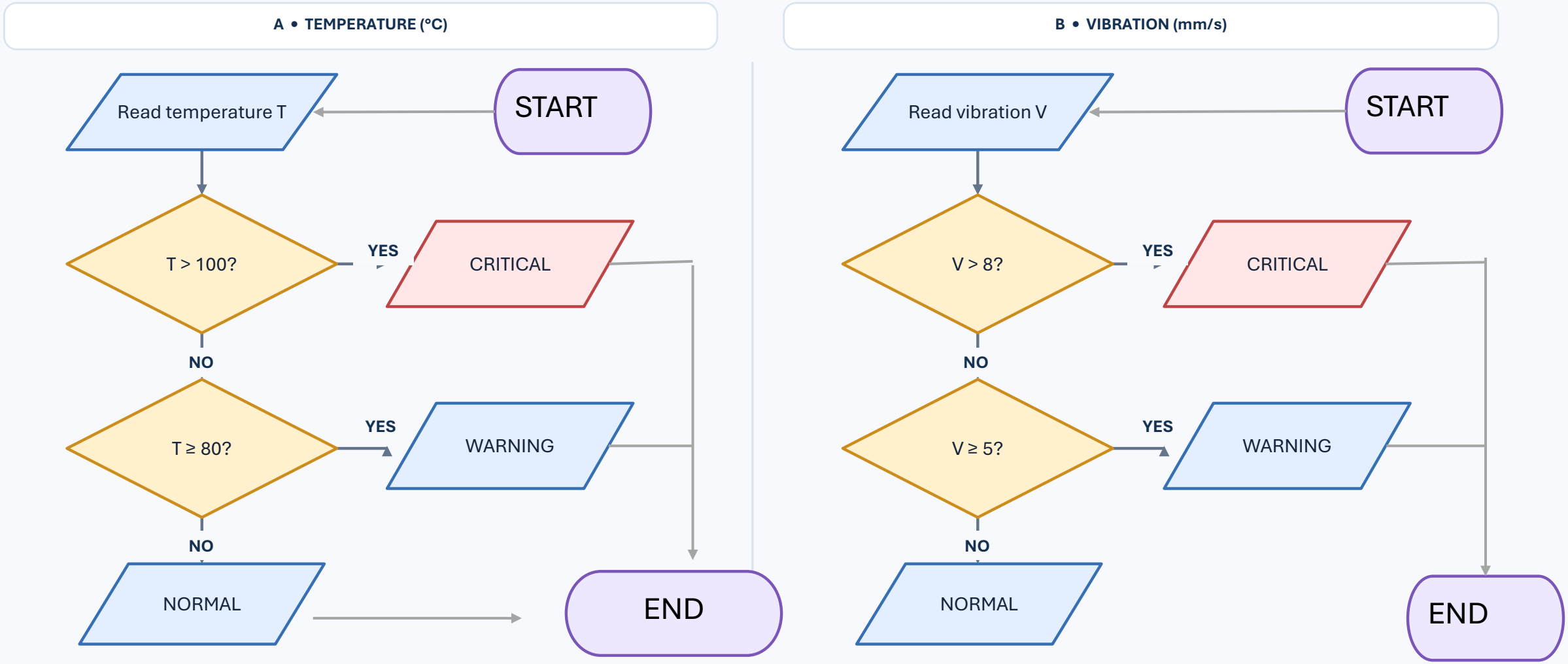
Equipment | Maintenance management

Equipment additions update a CSV; the maintenance view prioritises records for follow-up.



Condition monitoring | Threshold decisions

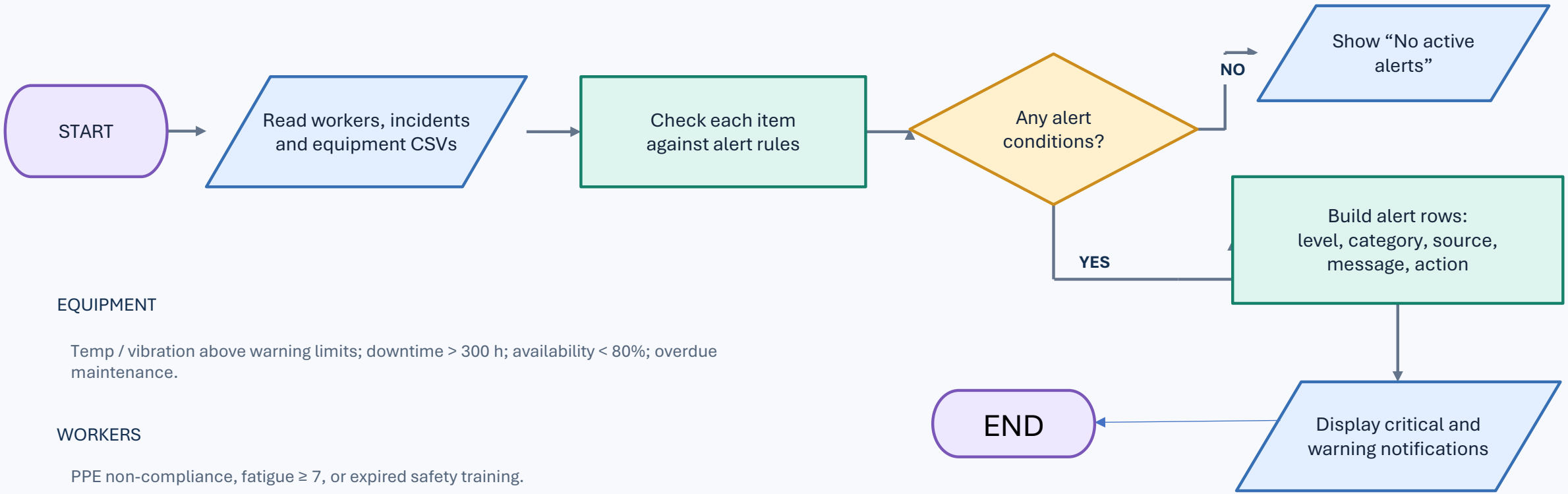
Educational prototype limits: temperature and vibration are classified separately; the most severe wins.



OVERALL STATUS: If either is CRITICAL → CRITICAL; otherwise if either is WARNING → WARNING; otherwise NORMAL.

Alerts | Generate safety and equipment warnings

The system checks three data sources, combines alert records, and displays the results.



EQUIPMENT

Temp / vibration above warning limits; downtime > 300 h; availability < 80%; overdue maintenance.

WORKERS

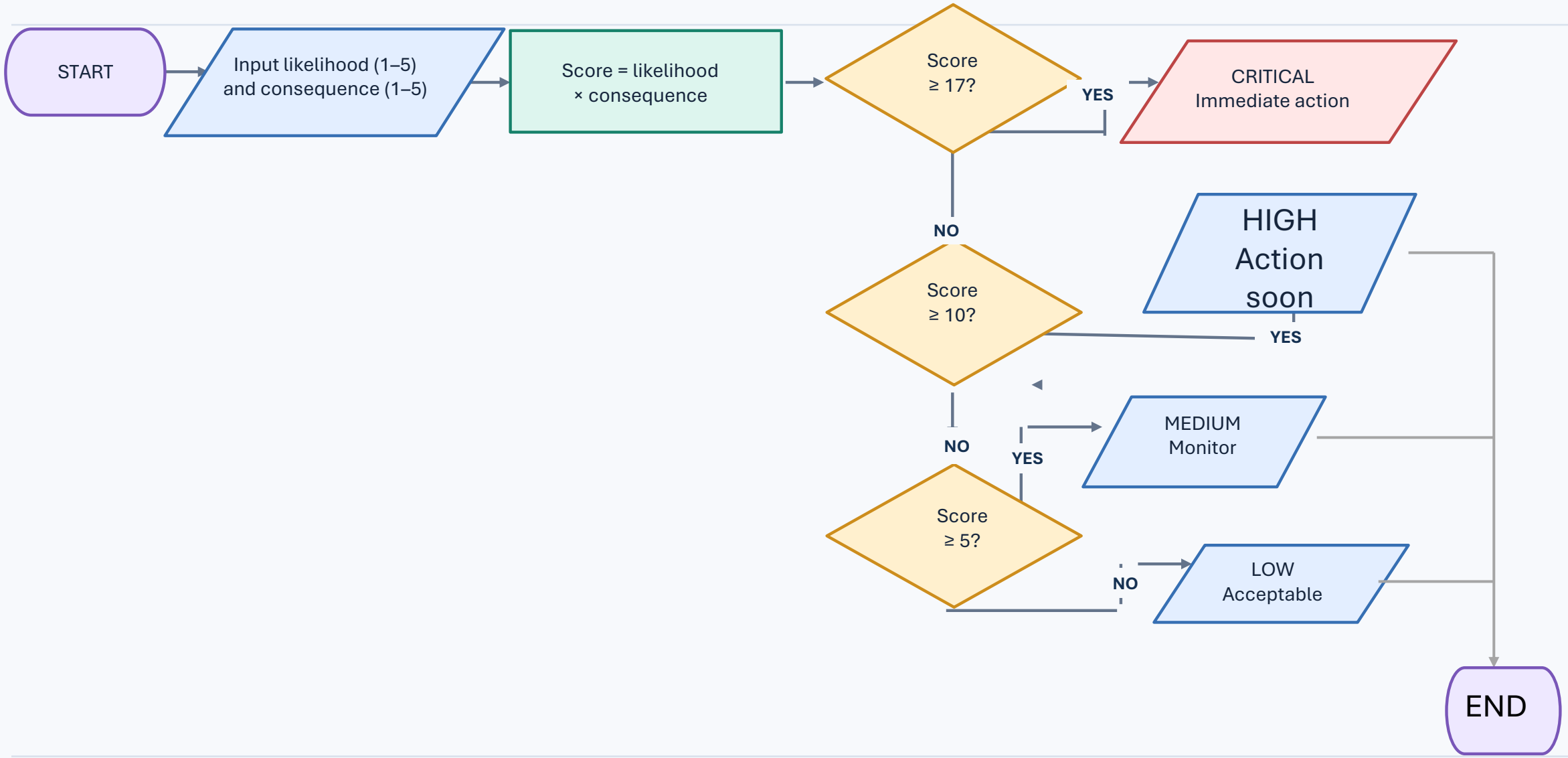
PPE non-compliance, fatigue ≥ 7, or expired safety training.

INCIDENTS

Severity is Critical and status is not Closed.

Risk assessment | Score classification

The risk score is the product of two inputs; sequential decisions determine its category.



Analytics | Reports | User administration

The last group of features handles business questions, data exports, and user accounts.

